EXPERIMENT 6

Bank Loan Application Automation Suite — Workflow Options, Deluge Scripting, Testing and Debugging

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Subject: Low Code Application Development using Zoho Creator
Project: Bank Loan Application & Pre-Screening Management

Aim

To build an automation layer on Loan_Application_Form using all eight workflow options of Zoho Creator (Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows and Functions), to write the supporting Deluge scripts, and to test and debug them.

Introduction

Capturing and displaying loan data is only the start. Acknowledging applications, chasing missing documents, escalating large loans, tracking stages, reminding borrowers of EMIs and recalculating figures all depend on people unless they are automated. Zoho Creator separates automation by trigger: **event-driven** (Form and Report Workflows), **time-driven** (Schedules and Batch Workflows), **human-in-the-loop** (Approvals and Blueprints), and **reusable** (Functions). Payments workflows connect to a payment connector.

Procedure

1. Open **Workflow → Form workflows**, create On_Application_Submit_Workflow with the trigger *Create → On success*, and add the Deluge sendmail action of Code 6.1.
2. Open **Schedules**, create Daily_Pending_Document_Reminder_Schedule to run daily at 10:00 AM with the script of Code 6.2.
3. Open **Approvals** and create HighValue_Loan_Approval for records whose Loan_Amount_Requested exceeds the threshold; name the credit-committee approvers and the quorum.
4. Open **Payments** and configure Loan_Disbursal_Payment_Workflow to start when Status changes to Approved, with three retries on the status check.
5. Open **Blueprints** and define Loan_Lifecycle_Blueprint with the stages and role-based transitions of Table 6.3; make the comment field mandatory.
6. Open **Batch Workflows** and create Bulk_EMI_Reminder_Batch to run daily at 7:00 AM in batches of 200 records.
7. Open the report workflow settings and create OnEdit_Recalculate_EMI_Workflow that fires only when amount or rate changes.
8. Create the custom function Generate_Sanction_Letter_Function and call it from an approval action and from a “Resend Sanction Letter” button.
9. Test each workflow with the plan of Table 6.4 and record failures in the debug log (Deluge *info* statements and the execution history).

Workflow Options Used and Their Mapping

Table 6.1: Eight workflow options

Option	Workflow name	Purpose	Trigger / frequency
Form Workflow	On_Application_Submit_Workflow	Acknowledgement to the applicant with the Loan Application ID; notify the loan officer	On Add of Loan_Application_Form
Schedule	Daily_Pending_Document_Reminder_Schedule	Remind applicants with pending KYC or income proof	Daily at 10:00 AM until documents are complete
Approval	HighValue_Loan_Approval	Route loans above the threshold to the credit committee	On Edit when amount exceeds the threshold
Payments	Loan_Disbursal_Payment_Workflow	Start the disbursal and update Disbursal_Status	When Status changes to Approved
Blueprint	Loan_Lifecycle_Blueprint	Enforce the loan stages with role-based transitions	Every record; Submitted → Document Verification → Credit Check → Approved → Disbursed / Rejected
Batch Workflow	Bulk_EMI_Reminder_Batch	EMI reminders to borrowers whose instalment is due within 3 days	Daily at 7:00 AM, batches of 200
Report Workflow	OnEdit_Recalculate_EMI_Workflow	Recalculate the EMI after a manual edit in the report	On Edit of a report record
Function	Generate_Sanction_Letter_Function	Reusable sanction-letter generation and e-mail	After approval, or by button

Deluge Scripts (reference implementations)

The scripts below are reference implementations written from the design. Field link names that were not created in Experiments 4 and 5 (Documents_Complete, Reminder_Count, Approved_Amount, Assigned_Officer_Email) are assumed supporting fields and have to be added to the form before the scripts are used.

```
// Code 6.1 — On_Application_Submit_Workflow (Form workflow, On Create > On Success)
sendmail
[
    from    : zoho.adminuserid
    to      : input.Email_Address
    subject : "Loan application received - " + input.Loan_Application_ID
    message : "Dear " + input.Applicant_Name.first_name + ",<br>We have received your " +
              input.Loan_Type + " application. Your reference number is " +
              input.Loan_Application_ID + ". A loan officer will call you at " +
              input.Preferred_Callback_Time + "."
]
sendmail
[
    from    : zoho.adminuserid
    to      : input.Assigned_Officer_Email
    subject : "New loan application " + input.Loan_Application_ID
    message : "A new " + input.Loan_Type + " application is ready for triage."
]
```

Code 6.1 — Acknowledgement and officer notification

```
// Code 6.2 — Daily_Pending_Document_Reminder_Schedule
pending = Loan_Application_Form[Documents_Complete == false && Reminder_Count < 5];
for each rec in pending
{
    sendmail
    [
        from    : zoho.adminuserid
        to      : rec.Email_Address
        subject : "Documents pending for " + rec.Loan_Application_ID
        message : "Please upload your KYC and income proof to continue your loan application."
    ]
    rec.Reminder_Count = rec.Reminder_Count + 1;
}
```

Code 6.2 — Pending-document reminder (stops after five reminders)

```
// Code 6.3 — Generate_Sanction_Letter_Function (custom function)
string Generate_Sanction_Letter_Function(int applicationId)
{
    rec = Loan_Application_Form[ID == applicationId];
    if (rec.Applicant_Name == null || rec.Approved_Amount == null)
    {
        return "Failed: applicant name or approved amount missing";
    }
    sendmail
    [
        from    : zoho.adminuserid
        to      : rec.Email_Address
        subject : "Loan sanction letter - " + rec.Loan_Application_ID
        message : "Dear " + rec.Applicant_Name.first_name + ", your loan of " + rec.Approved_Amount +
                  " has been sanctioned. Estimated EMI: " + rec.Estimated_Monthly_EMI + "."
    ]
    return "Sent";
}
```

Code 6.3 — Reusable function with input validation

Governance Rules

Table 6.2: Configuration and governance rules

Workflow	Rule
Form workflow	Fails gracefully and logs an error if the e-mail is invalid, instead of blocking submission
Schedule	Maximum 5 reminders per application, then manual follow-up call
Approval	Approver group with a minimum quorum
Payments	Three retries of the status-check callback, then flagged for the finance team
Blueprint	Mandatory comment on every stage transition
Batch	Batch size configurable; default 200 records
Report workflow	Fires only when Loan_Amount_Requested or Interest_Rate_Offered changes
Function	Validates that applicant name and approved amount exist before running

Blueprint Stages

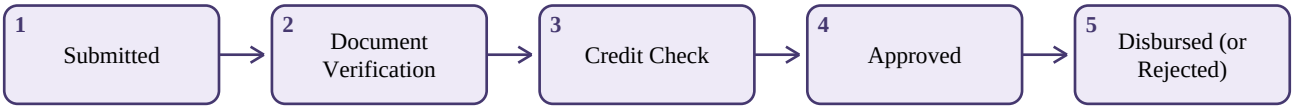


Fig 6.1 — Loan_Lifecycle_Blueprint stages (schematic drawn from the specification; not a screenshot)

Table 6.3: Suggested role-based transitions (assumed; align with the roles created in Experiment 8)

Transition	Who may perform it
Submitted → Document Verification	Loan Officer
Document Verification → Credit Check	Loan Officer
Credit Check → Approved	Credit Committee / Branch Manager
Approved → Disbursed	Finance Team
Any stage → Rejected	Loan Officer or Branch Manager (comment required)

Testing and Debugging Plan

Table 6.4: Test cases (to be executed and results recorded)

Test	Action	Expected result
T1 Acknowledgement	Submit a test application	Applicant receives the e-mail with the reference number
T2 Invalid e-mail	Submit with a malformed address	Submission succeeds; error is logged
T3 Schedule	Run the schedule manually with one pending record	Reminder sent; Reminder_Count increases by 1
T4 Reminder cap	Set Reminder_Count to 5	No further reminder is sent
T5 Approval	Edit amount above the threshold	Status shows Pending Committee Approval
T6 Blueprint	Try to move a record directly to Disbursed	Transition is refused
T7 EMI recalculation	Edit the amount in the report	Estimated EMI changes immediately
T8 Function	Call with a record that has no approved amount	Returns the failure message and sends no e-mail

Debugging aids used: **info** statements printed to the execution log, the workflow execution history, and checking that link names in scripts match the field names exactly. Typical faults are a misspelt link name, a missing supporting field, and a criterion that never matches because of a data-type difference.

Output

SCREENSHOT TO ATTACH

Configuration screens of the form workflow, schedule, approval, payments, batch and report workflows

Fig 6.2 — Workflow Builder screens

SCREENSHOT TO ATTACH

Loan_Lifecycle_Blueprint stage diagram with transition permissions

Fig 6.3 — Blueprint stage diagram

SCREENSHOT TO ATTACH

Acknowledgement e-mail received after submitting a test application

Fig 6.4 — Acknowledgement e-mail

SCREENSHOT TO ATTACH

Generate_Sanction_Letter_Function script and its execution called from two different workflow types

Fig 6.5 — Function and its two call sites

Result

Eight automations, one for each workflow option, were configured for the loan process, with reference Deluge scripts, governance rules and a test plan for verification.

Conclusion

Event-driven workflows react at once, schedules and batches run on the clock, approvals and blueprints put people and permissions into the process, and functions keep shared logic in one place. Testing each workflow with a normal case and a failing case, and reading the execution log, is what turns a configured workflow into one that can be trusted.